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EVOHEAT H8A INSTALLATION AND INSTRUCTION MANUAL

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Hero Series Manual

Inverter Air Source Water Heat Pump

Floor heating and Air-con Unit

Installation and Instruction Manual

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Preface

- In order to provide the customers with high quality, strong reliability and good versatility product, this heat pump is produced by strict design and manufacture standards.
  This manual includes all the necessary information about installation, debugging, discharging and maintenance. Please read this manual carefully before you open or maintain the unit.
  The manufacture of this product will not be held responsible if someone is injured or the unit is damaged, as a result of improper installation, debugging, unnecessary maintenance which is not in line with this manual.
  The unit must be installed by qualified personnel.
- It is vital that the below instructions are adhered to at all times to keep the warranty.
  —The unit can only be opened or repaired by qualified installer or an authorised dealer.
  —Maintenance and operation must be carried out according to the recommended time and frequency, as stated in this manual.
  —Use genuine standard spare parts only.
  Failure to comply with these recommendations will invalidate the warranty.
- Inverter air source water heat pump is a kind of high efficiency, energy saving and environment friendly equipment, which is mainly used for house warming. It can work with any kind of indoor unit such fan coil, radiator, or floor heating pipe, by provide warm or hot water. One unit of monobloc heat pump can also work with several indoor units.
  The air source water heat pump unit is designed to have heat recovery by using super heater which can provide hot water for sanitary purpose.

This series of heat pump unit owns following features:

1 Advanced controlling
The PC microcomputer based controller is available for the users to review or set the running parameters of the heat pump. Centralized controlling system can control several units by PC.

2 Nice appearance
The heat pump is designed with beautiful looking. The monobloc one has the water pump included which is very easy for installation.

3 Flexible installation
The unit has smart structure with compact body, just simple outdoor installation is needed.

4 Quiet running
High quality and efficient compressor, fan and water pump is used to ensure the low noise level with insulation.

5 Good heat exchange rate
The heat pump unit use special designed heat exchanger to enhance whole efficiency.

6 Large working range
This series of heat pump is designed to work under different working conditions as low as -15 degrees for heating.

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To prevent the users and others from the harm of this unit, and avoid damage on the unit or other property, and use the heat pump properly, please read this manual carefully and understand the following information correctly.

Mark Notes

Mark	Meaning
 WARNING	A wrong operation may lead to death or heavy injury on people.
 ATTENTION	A wrong operation may lead to harm on people or loss of material.

Icon notes

Icon	Meaning
	Prohibition. What is prohibited will be nearby this icon
	Compulsory implement. The listed action need to be taken.
	**ATTENTION** (include **WARNING**) Please pay attention to what is indicated.

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Warning

Installation	Meaning
 Professional installer is required.	The heat pump must be installed by qualified personals, to avoid improper installation which can lead to water leakage, electrical shock or fire.
 Earthing is required	Please make sure that the unit and power connection have good earthing, otherwise may cause electrical shock.

Operation	Meaning
 PROHIBITION	DO NOT put fingers or others into the fans and evaporator of the unit, otherwise harm may be occurred.
 Shut off the power	When there is something wrong or strange smell, the power supply need to be shut off to stop the unit. Continue to run may cause electrical short or fire.

Move and repair	Meaning
 Entrust	When the heat pump need to be moved or installed again, please entrust dealer or qualified person to carry it out. Improper installation will lead to water leakage, electrical shock, injury or fire.
 Entrust	It is prohibited to repair the unit by the user himself, otherwise electrical shock or fire may be occur.
 Prohibit	When the heat pump need to be repaired, please entrust dealer or qualified person to carry it out. Improper movement or repair on the unit will lead to water leakage, electrical shock, injury or fire.

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ATTENTION

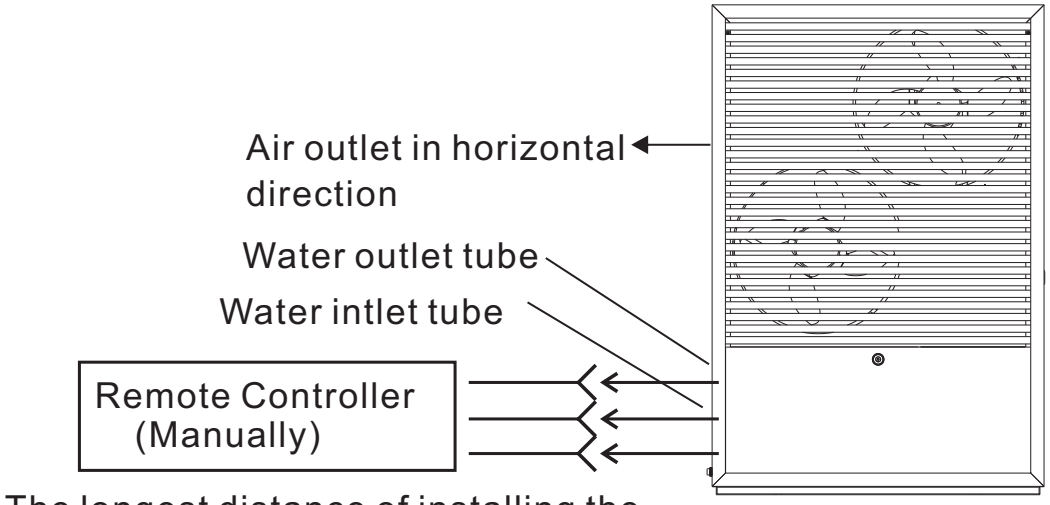
Installation	Meaning
 Installation Place	The unit CANNOT be installed near the flammable gas. Once there is any leakage of the gas, fire can be occur.
 Fix the unit	Make sure that the basement of the heat pump is strong enough, to avoid any decline or fall down of the unit
 Need circuit breaker	Make sure that there is circuit breaker for the unit, lack of circuit breaker can lead to electrical shock or fire.

Operation	Meaning
 Check the installation basement	Please check the installation basement in a period (one month), to avoid any decline or damage on the basement, which may hurt people or damage the unit
 Switch off the power	Please switch off the power for clean or maintenance.
 Prohibition	It is prohibited to use copper or iron as fuse. The right fuse must be fixed by electrician for the heat pump.
 Prohibition	It is prohibited to spray the flammable gas to the heat pump, as it may cause fire.

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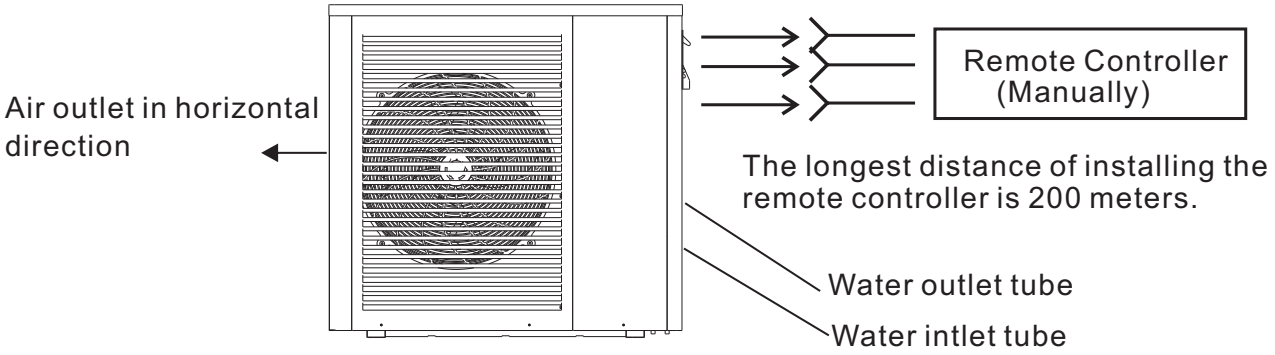
1. Appearance and Structure of the heat pump

H15A



The longest distance of installing the remote controller is 200 meters.

H8A



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2. The data of unit

Model		H15A	H8A
Cooling Capacity	kW	10	7
	BTU/h	34000	23800
Heating Capacity	kW	15	9
	BTU/h	51000	30600
Electric Heating Power	kW	3.0	3.0
Cooling Power Input	kW	3.1	2.6
Heating Power Input	kW	3.1	2.6
Operation Electric Current(cooling/heating)	A	14.1/14.1	11.8/11.8
Cooling Capacity Range	kW	6.0～15.0	2.0～10
Heating Capacity Range	kW	6.0～18.0	2.5～11.5
Cooling Power Input Range	kW	1.6～6.0	1.0～3.4
Heating Power Input Range	kW	1.6～7.0	0.8～3.4
Power Supply		230V～/50Hz	230V～/50Hz
Compressor Quantity		1	1
Compressor Model		ScrolII	Rotary
Fan Quantity		2	1
Fan Power Input	W	110×2	110
Fan Rotate Speed	RPM	850	850
Noise	dB(A)	58	54
Water Pump Input	kW	0.18	0.18
Max water head	m	12.5	12.5
Water Connection	inch	1 1/4	1
Water Flow Volume	m^3/h	2.5	1.6
External Pressure	m	10	10
Unit Net Dimensions (L/W/H)	mm	1000*465*1475	980*465*900
Unit Shipping Dimensions (L/W/H)	mm	1050*510*1500	1010*490*920
Net Weight	kg	See nameplate	See nameplate
Shipping Weight	kg	See package label	See package label

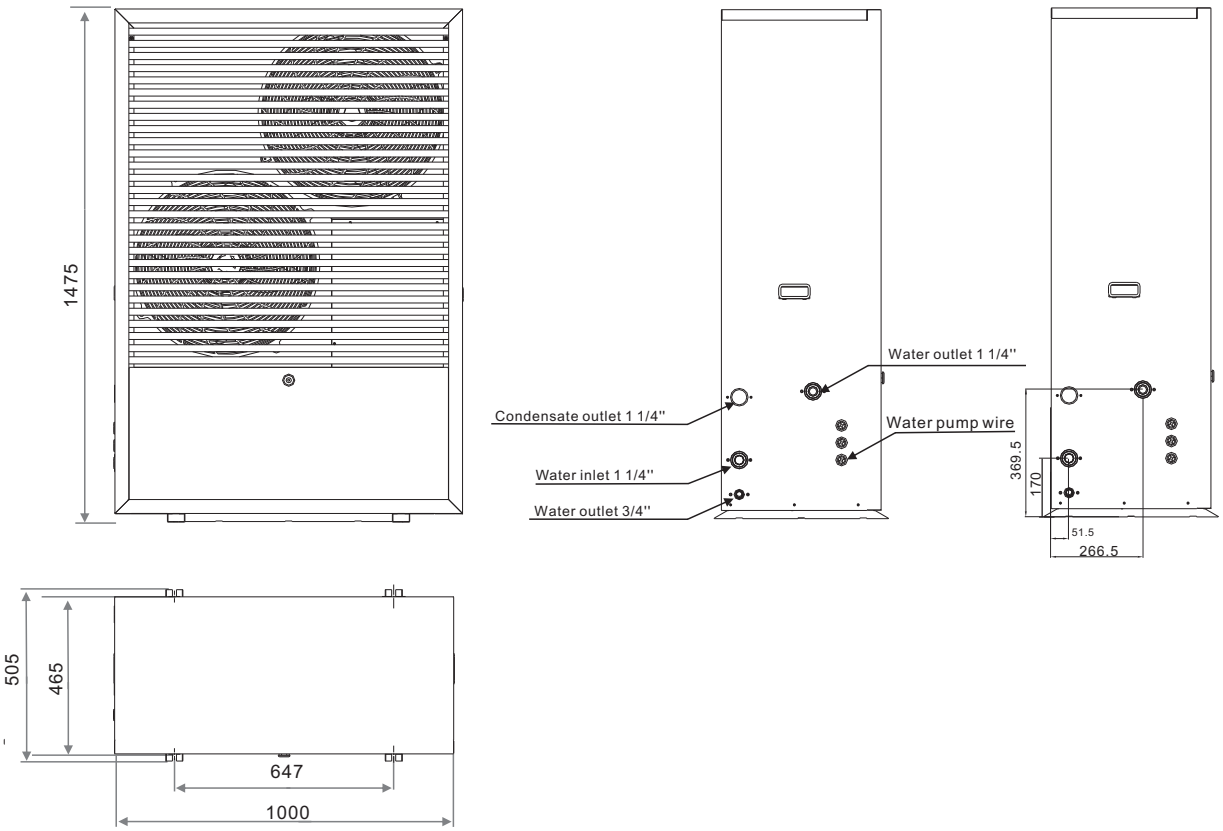
Cooling working condition:(DB/WB)35℃/24℃, (Outlet/Inlet) 7℃/12℃.
Heating working condition: (DB/WB) 7℃/6℃. (Outlet/Inlet) 35℃/30℃.

BS EN 14511-1-2013 Air conditioner, whole liquid cooling machine, electric compressor.
Part2: Test condition Part3:Test method Part4:related requirements.

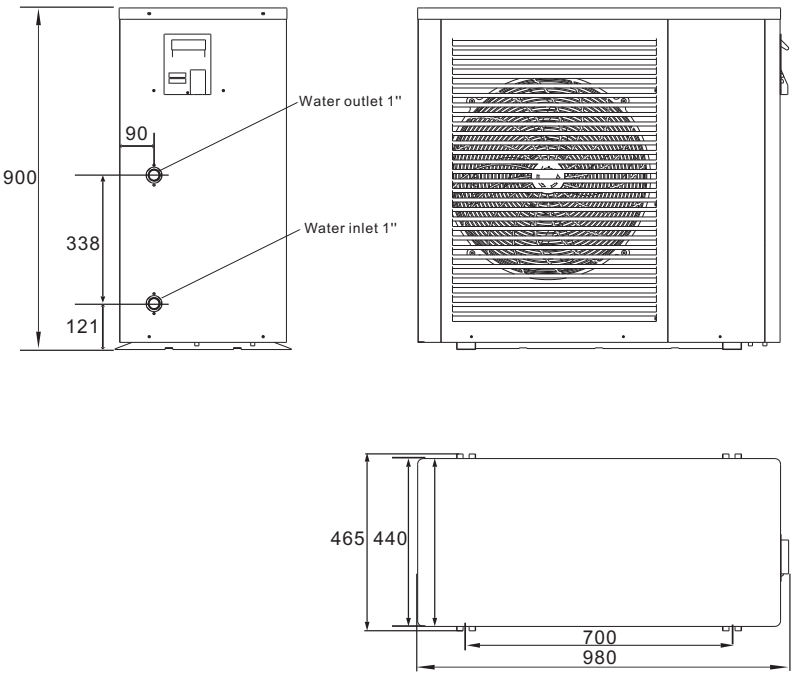
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3 Unit dimension

Models: H15A



Models:H8A



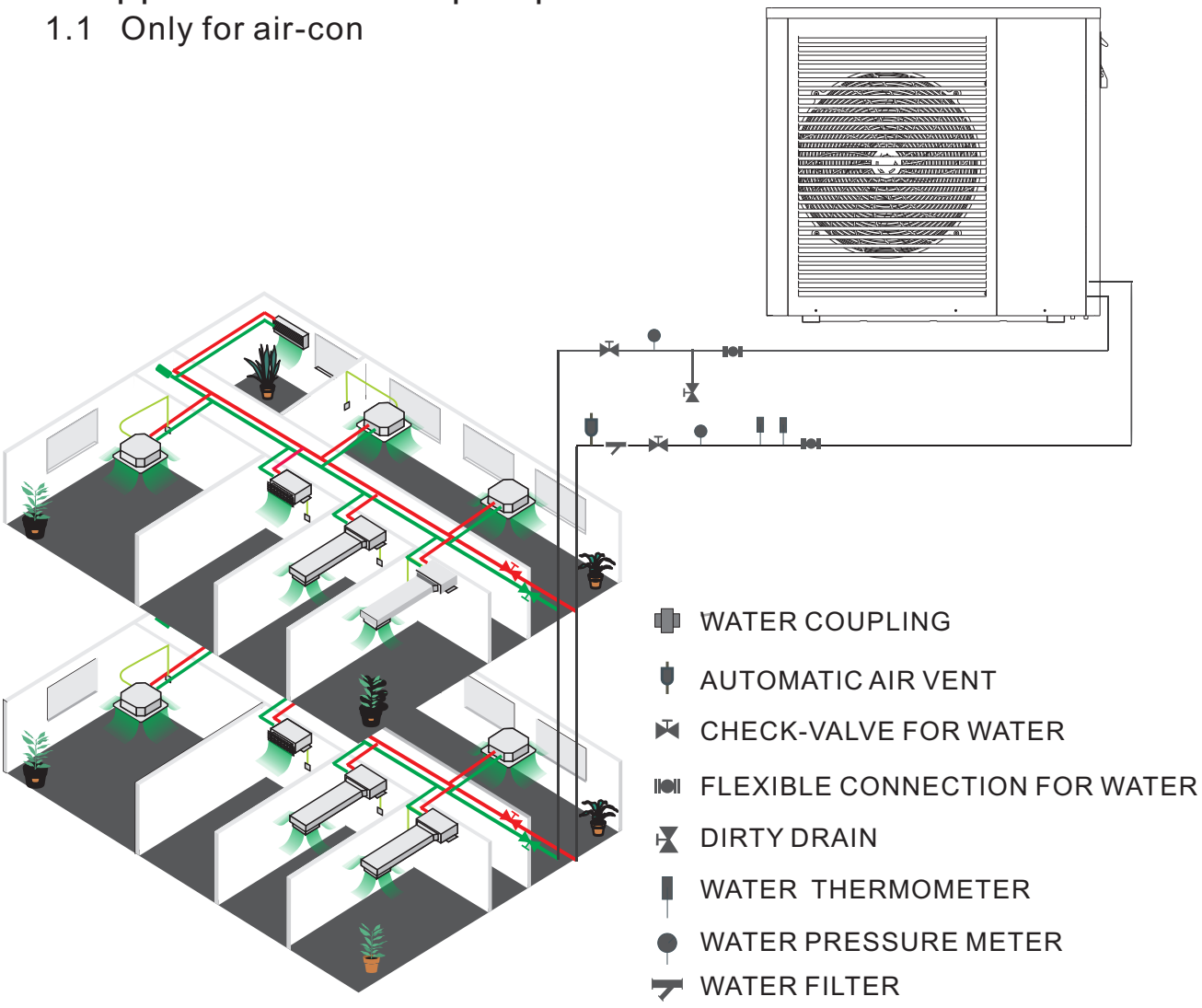
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Unit features

1. Plate heat exchanger
   Use the SWEP efficient heat exchanger with small size and high efficiency.
2. Environmentally friendly refrigerant
   Use the new generation of environmentally friendly refrigerant R410, which is harmless to the ozone sphere.
3. Heating in frigid environment.
   Optimized designed unit can achieve the heating function normally even when the ambient temperature is -15℃.
4. Spraying condensation
   The water-cooling system will be opened automatically to guarantee the normal operation of cooling when the ambient temperature is higher than 30℃.

1 Application of heat pump

1.1 Only for air-con



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2 Choose a right heat pump unit

2.1 Based on the local climate condition, construction features and insulation level, calculate the required cooling(heating) capacity per square meter.

2.2 Conclude the total capacity which will be needed by the construction.

2.3 According to the total capacity needed, choose the right model by consulting the heat. pump features as below:

- Heat pump features

  Cooling only unit： chilled water outlet temp. at 5-15℃, maximum ambient temp. at 43℃.

  Heating and Cooling unit：for cooling chilled water outlet temp. at 5-15℃,maximum ambient temp. at 43℃. For heating, warm water inlet temp. at 40-50℃, minimum ambient temp. at -10℃.

- Unit application

  Inverter air source water heat pump is used for house, office, hotel, and so forth, which need heating or cooling separately, with each area need to be controlled.

3 Installation place

- The unit can be installed on any place outdoor which can carry heavy machine such as terrace, housetop, ground and so on.
- The location must have good ventilation.
- The place is free from heat radiation and other fire flame.
- A pall is needed in winter to protect the heat pump from snow.
- There must be not obstacles near the air inlet and outlet of the heat pump.
- A place which is free from strong air blowing.
- There must be water channel around the heat pump to drain the condensing water .
- There must be enough space around the unit for maintenance.

4 Installation method

The heat pump can be installed onto the concrete basement by expansion screws, or onto a steel frame with rubber feet which can be placed on the ground or housetop. Make sure that the unit is placed horizontally.

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5 Water loop connection

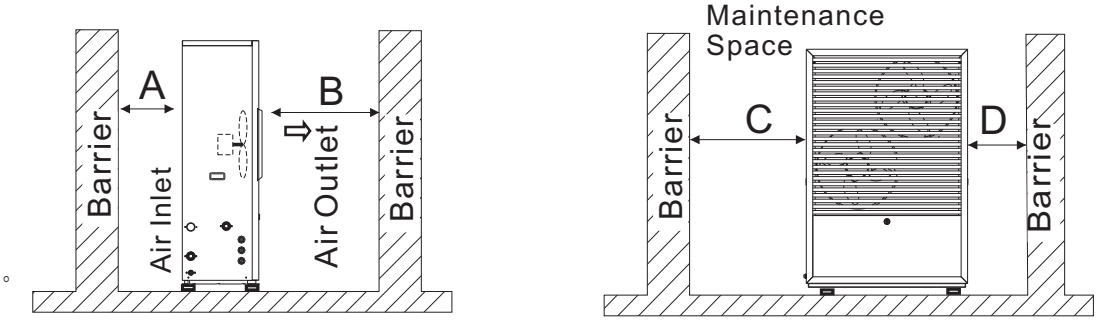
Please pay attention to below matters when the water pipe is connected:

- Try to reduce the resistance to the water from the piping.
- The piping must be clear and free from dirty and blocks. Water leakage test must be carried out to ensure there is no water leaking. And then the insulation can be made.
- Attention that the pipe must be tested by pressure separately. DO NOT test it together with the heat pump.
- There must be expansion tank on the top point of the water loop, and the water level in the tank must be at least 0.5 meter higher than the top point of the water loop.
- The flow switch is installed inside of the heat pump, check to ensure that the wiring and action of the switch is normal and controlled by the controller.
- Try to avoid air stayed inside of the water pipe, and there must be air vent on the top point of the water loop.
- There must be thermometer and pressure meter at the water inlet and outlet, for easy inspection during running.

6 Power supply connection

- Open the front panel, and open the power supply access.
- The power supply must go through the wire access and be connected to the power supply terminals in the controlling box. Then connect the 3-signal wire plugs of the wire controller and main controller.
- If the outside water pump is needed, please insert the power supply wire into the wire access also and connect to the water pump terminals.
- If an additional auxiliary heater is need to be controlled by the heat pump controller, the relay (or power) of the aux-heater must be connected to the relevant output of the controller.

7 Location of the unit



The picture shows the location of horizontal air outlet unit.

Attention

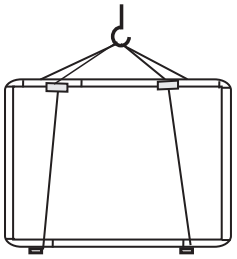
Requirements
A>500mm ; B>1500mm ;
C>1000mm ; D>500mm

The minimum ventilation distance in diagram 1.

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8 Transit

When the unit need to be hung up during installation, a 8 meters cable is needed, and there must be soft material between the cable and the unit to prevent damage to the heat pump cabinet. (See picture 1)



Picture 1

WARNING

DO NOT touch the heat exchanger of the heat pump with fingers or other objects!

9 Trial Running

Inspection before trial running

- Check the indoor unit, and make sure that the pipe connection is right and the relevant valves are open .
- Check the water loop, to ensure that the water inside of the expansion tank is enough, the water supply is good, the water loop is full of water and without any air. Also make sure there is good insulation for the water pipe.
- Check the electrical wiring. Make sure that the power voltage is normal, the screws are fastened, the wiring is made in line with the diagram, and the earthing is connected.
- Check the heat pump unit including all of the screws and parts of the heat pump to see if they are in good order. When power on, review the indicator on the controller to see if there is any failure indication. The gas gauge can be connected to the check valve to see the high pressure(or low pressure) of the system during trial running.

Trial running

- Start the heat pump by press " ↵ " key on the controller. Check whether the water pump is running, if it runs normally there will be 0.2 MPa on the water pressure meter.
- When the water pump runs for 1 minutes, the compressor will start. Hear whether there is strange sound from the compressor. If abnormal sound occurs please stop the unit and check the compressor. If the compressor runs well please look for the pressure meter of the refrigerant.
- Then check whether the power input and running current is in line with the manual. If not please stop and check.
- Adjust the valves on the water loop, to make sure that the hot(cool) water supply to each door is good and meet the requirement of heating(or cooling).
- Review whether the outlet water temperature is stable.
- The parameters of the controller are set by the factory, it is not allowed to change then by user himself.

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Usage

1.Interface



2.Function and Display Icon

NO.	Icon	Key name	Key functions
①		Mode	Long press "Mode", to switch between cooling and heating
②		Fan Speed	Short press "Fan Speed", to enter the Query interface
③		On/Off	In OFF state, long press "ON/OFF", to enter the Startup interface In ON state, long press "ON/OFF", to enter the Shutdown interface
④		Up	Short press "UP", to modify the temperature or clock
⑤		Down	Short press "DOWN", to modify the temperature or clock
⑥		Cooling	In the state of cooling, the icon Cool is on
⑦		Heating	In the state of heating, the icon Hot is on
⑧		Def	In the state of defrosting, the icon Def is on

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NO.	Icon	Key name	Key functions
⑨		Inlet	Inlet icon
⑩		Outlet	Outlet icon
⑪		Compressor	When the compressor starts up, in the main display area, thecompressor status icon is on, and when the compressor shuts down, the icon is off.
⑫		Pump	When the Pump starts up, in the main display area, the Pump status icon is on, and when the Pump shuts down, the icon is off.
⑬		Fan	When the Fan starts up, in the main display area, the Fan status icon is on, and when the Fan shuts down, the icon is off.
⑭		Electric Heating	When the Electric Heating starts up, in the main display area, the Electric Heating status icon is on, and when the Electric Heating shuts down, the icon is off.
⑮		4-way valve	When the 4-way valve starts up, in the main display area, the 4-way valve status icon is on, and when the 4-way valve shuts down, the icon is off.

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3.Usage Of Wire Controller

3.1 Startup & Shutdown



Note: the Startup & Shutdown operation can only be performed in the main interface.

Note: 1) Without Thermostat, wired controller can be used for Startup & Shutdown;
2) With Thermostat, wired controller can only be used for Shutdown, and cannot be used for Startup;
3) When using the remote control, wired controller is shown as below:

Startup Interface



Shutdown Interface



3.2 Mode

In the main interface, long press for " M " second, to switch between the cooling and heating modes.



Cooling Interface



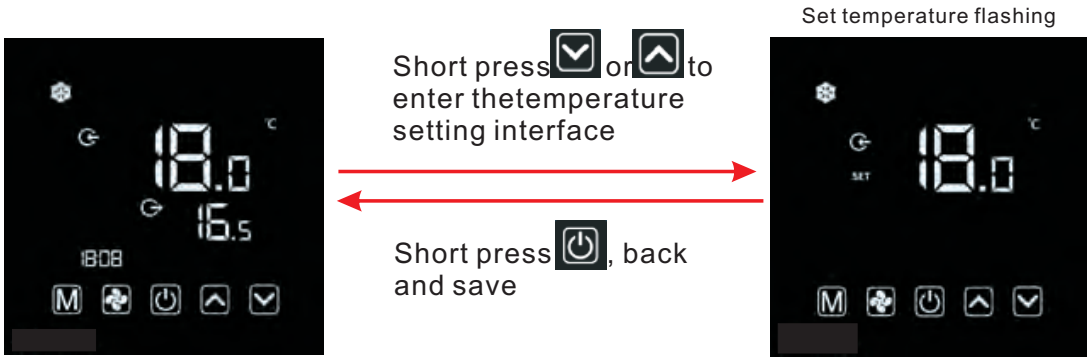
Heating Interface



Def Interface
(heating mode only)

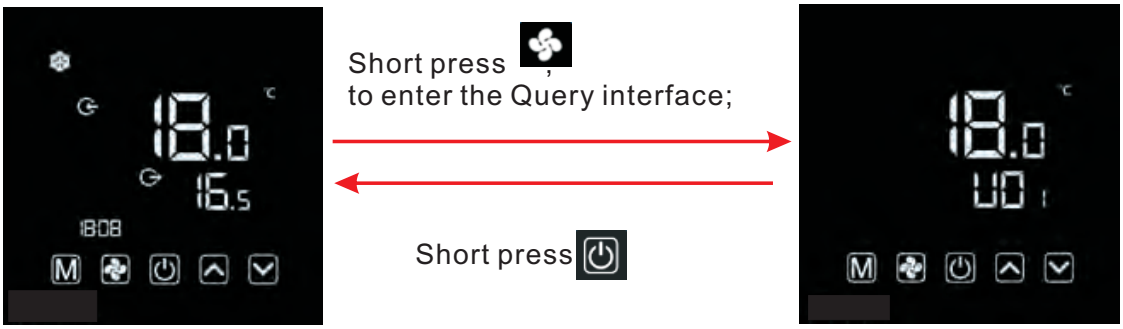
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3.3 Temperature



Note: In the temperature setting interface, for 20 seconds without operation, the system will automatically record the setting, and back to the main interface.

3.4 State parameter query



Note: In the query parameter interface, it cannot automatically back to the main interface.

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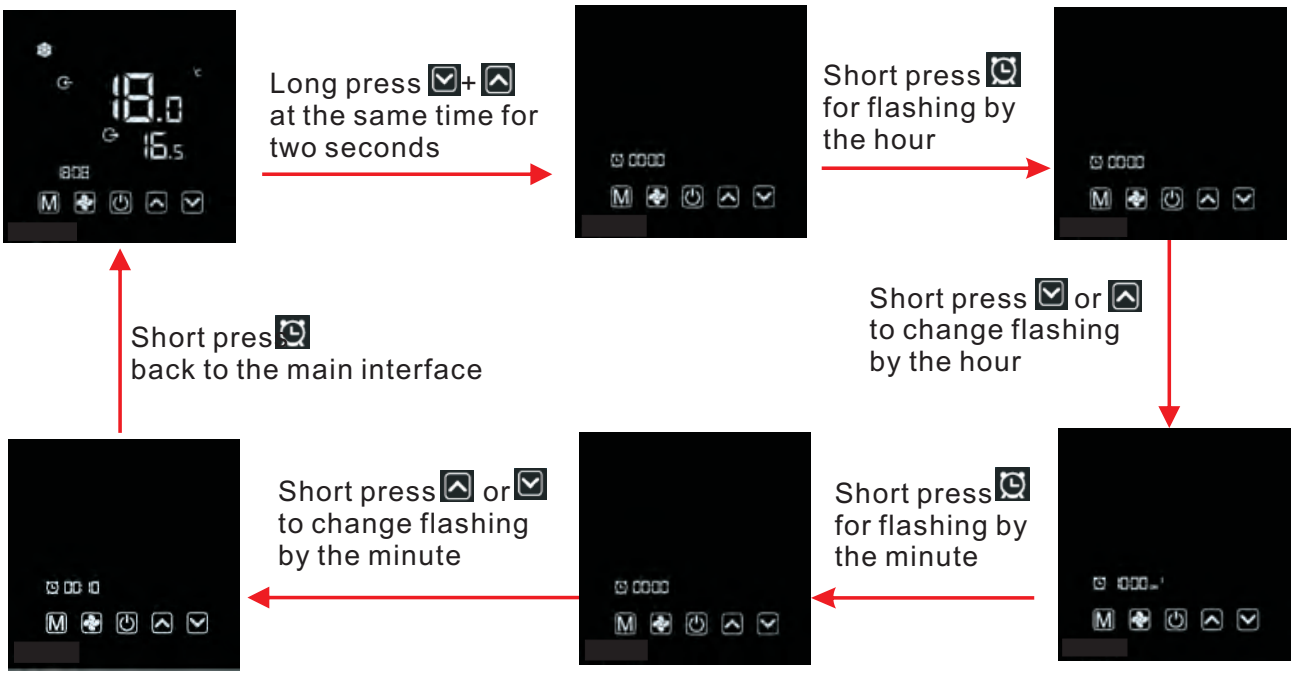
3.5 State query list:

State No	State Name
U01	Inlet Temperature
U02	Outlet Temperature
U03	Suction Temperature
U04	Discharge Temperature
U05	Ambient Temperature
U06	Coil Temperature
U07	Low Pressure
U08	High Pressure
U09	Electronic Expansion Valve Steps
U10	Compressor Actual Frequency
U11	Compressor Setting Frequency
U12	Suction gas Superheating/10
U13	Discharge gas Superheating
U14	Fan 1 Set Speed
U15	Fan 2 Set Speed

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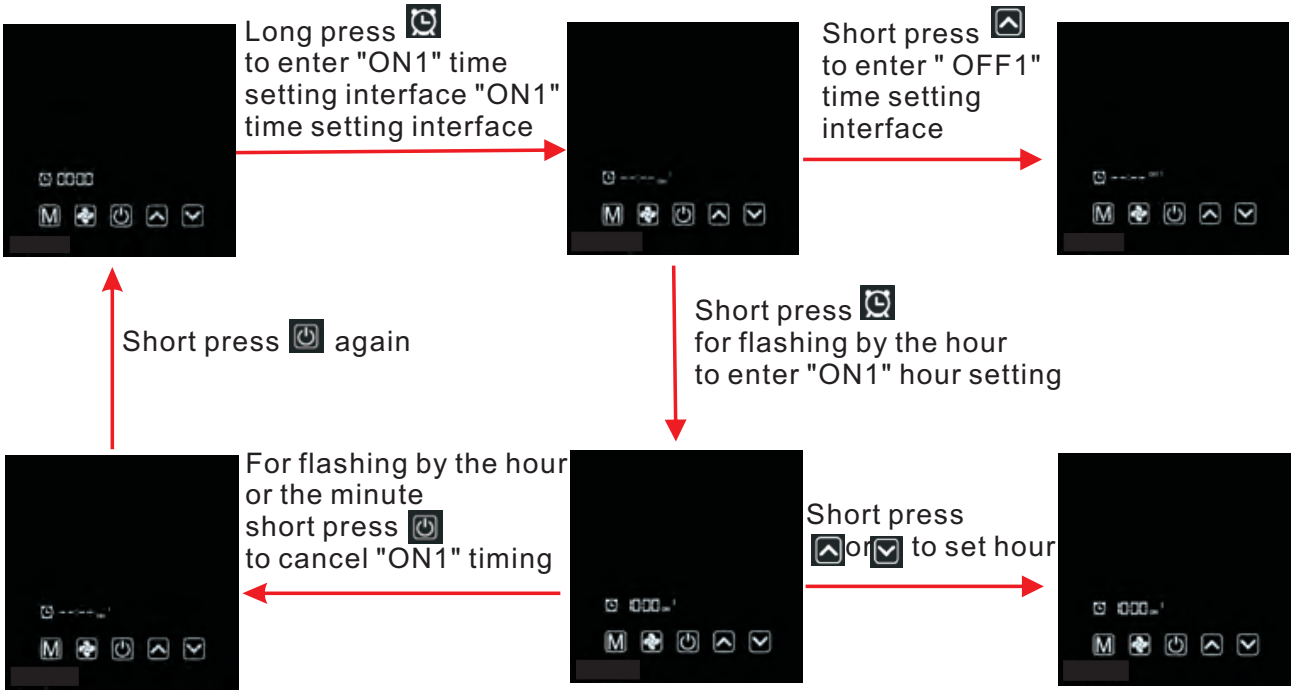
3.6 Clock Setting

3.6.1 Time Setting



Note: In the clock setting interface, for 20 seconds without operation, the system will automatically record the setting, and back to the main interface.

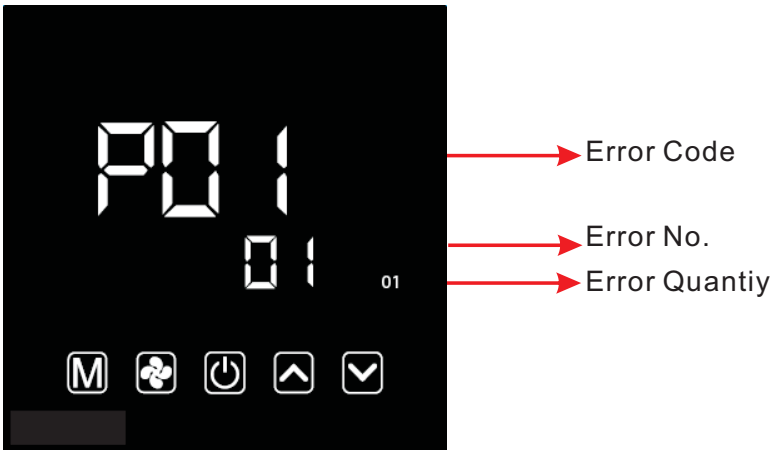
3.6.2 Set and cancel the timing
(if connected to the thermostat, the timing function is not available)



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4 Error Interface

4.1 Interface



Note:

1) .On the Error Interface, short press "ON/OFF", back to the main interface, and short press "UP" or "DOWN" to display multiple errors.

2).On the main interface, short press "ON/OFF" to back to the Error Interface. For 10 seconds without operation, automatically back to the Error Interface.

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5 Electronic Control Error Table

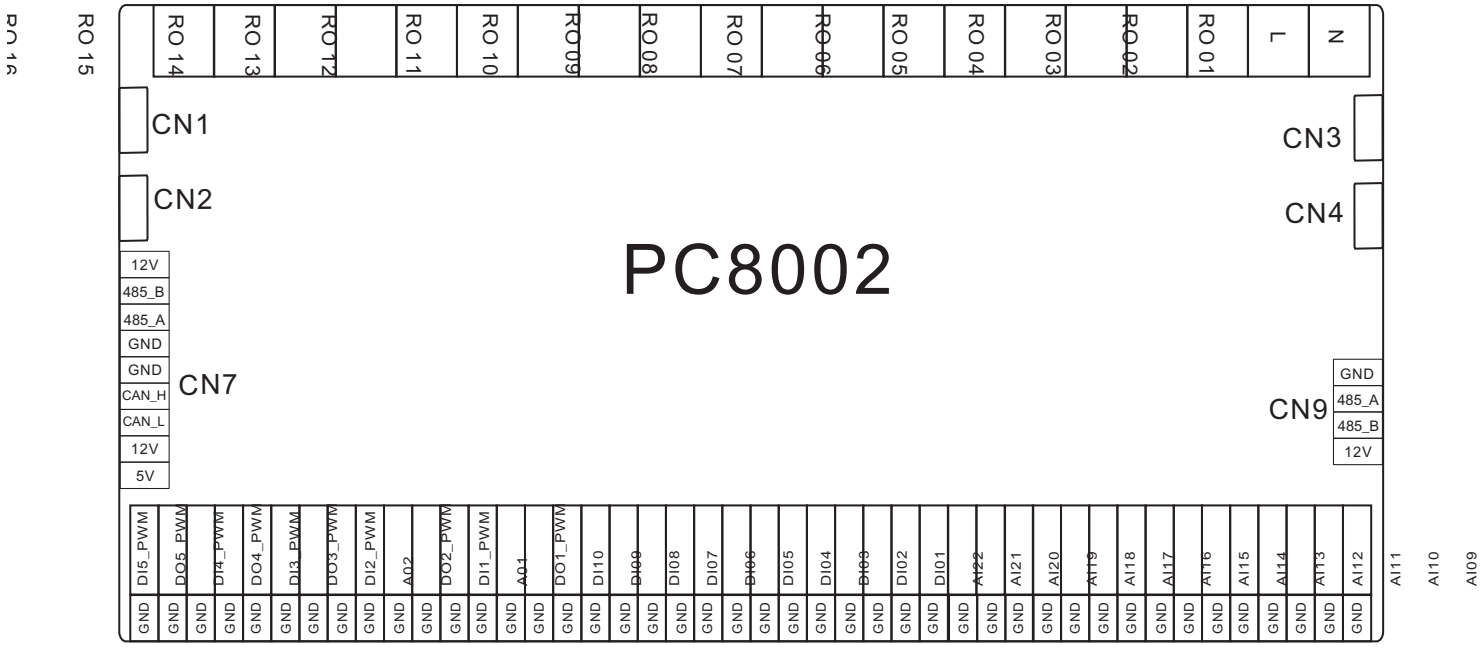
We can use Error Code to find the cause and the solution.

Error Code	Protection/Error
None	Standby
None	Normal Startup
P01	Inlet Temperature Error
P02	Outlet Temperature Error
P04	Ambient Temperature
P07	Suction Temperature Error
P08	Discharge Temperature Error
PP1	Low Pressure Sensor Erro
PP2	High Pressure Sensor Erro
E01	High Pressure Protection
E02	Low Pressure Protection
E03	Flow Switch Protection
E19	Winter Def Protection I
E20	Winter Def Protection II
E07	Def Protection
P81	Dischange High Temperature Protection
E44	Pump Working Environment Protection
E21	AC Input low voltage protection
E22	AC Input over-current protection shutdown
E23 (blank)	AC Input over-current protection alarm
E24	DC bus over-voltage protection
E25	DC bus low-voltage protection
E27	IPM over-current protection
E28	IPM overheating protection
E29	Radiator temperature sensor error
E30	Power component overheating protection shutdown
E31 (blank)	Power component overheating protection alarm
E32	IPM current sampling
E33	Compressor over-current protection shutdown
E34 (blank)	Compressor over-current protection alarm
E35	Input voltage phase
E36	Compressor start-up error
E37	Communication error between DSP and communication board
E38	Communication error between DSP with PFC
E39	Input voltage sampling
E40	EEPROM Error
E41	Compressor weak magnetic protection
E45	PFC Error
EE 8	Communication error between drive and main board
E08	Communication error between main board and wired controller

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6 Controller interfaces and definitions

6.1 Main board interfaces



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NO.	Main board	Instruction
1	N	Power Null Wire
2	L	Power Live Wire
3	RO 01	Not Used
4	RO 02	Pump
5	RO 03	4-way valve
6	RO 04	Fan
7	RO 05	Not Used
8	RO 06	Def Heating
9	RO 07	Crankshaft Heating
10	RO 08	Spray Valve
11	RO 09	Auxiliary Electric Heating
12	RO 10-RO 18	Not Used
13	AI 01	Inlet Temperature Sensor
14	AI 02	Outlet Temperature Sensor
15	AI 03	Discharge Temperature Sensor
16	AI 04	Not Used
17	AI 05	Suction Temperature Sensor
18	AI 06	Coil Temperature Sensor
19	AI 07	Environment Temperature Sensor
20	AI 08-AI 20	Not Used
21	AI 21	Suction pressure input
22	AI 22	Discharge pressure input

NO.	Main board	Instruction
23	DI 01	High voltage switch input
24	DI 02	Low voltage switch input
25	DI 03	Flow switch input
26	DI 04	Emergency switch input
27	DI 05	Mode Switch
28	DI 06	Master and slave switch
29	DI 07	Electric heating overload protection switch
30	DI 08-DI 10	Not Used
31	A 01-A 02	Not Used
32	D01-D02_PWM	Not Used
33	DI0-DI2_PWM	Not Used
34	12V	Wired Controller 12V
35	485-B	Wired Controller 485-B
36	485-A	Wired Controller 485-A
37	GND	Wired Controller GND
38	CAN-L	Not Used
39	CAN-H	Not Used
40	CN1	Electronic Expansion Valve A
41	CN2-CN4	Not Used

AI08 AI07 AI06 AI05 AI04 AI03 AI02 AI01

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A. Form of Troubleshooting

Protection/fault	Fault display	Cause	Way of troubleshooting
Standby	NO		
Normal booting	NO		
Water inlet temp. sensor error	P01	Water inlet temp sensor is in open circuit or short circuit	Check and replace temp. sensor
Water inlet temp.sensor error	P02	Water outlet temp sensor is in open circuit or short circuit	Check and replace temp. sensor
Ambient temp. sensor error	P04	Ambient temp sensor is in open circuit or short circuit	Check and replace temp. sensor
Suction temp. sensor error	P07	Suction temp. sensor is in open circuit or short circuit	Check and replace temp. sensor
Exhaust temp. sensor error	P08	Exhaust temp. Sensor is in open circuit or short circuit	Check and replace temp. sensor
Suction pressure sensor error	PP1	Suction pressure sensor is in open circuit or short circuit	Check and replace suction pressure sensor
Exhaust pressure sensor error	PP2	Exhaust pressure sensor is in open circuit or short circuit	Check and replace exhaust pressure sensor
High-pressure protection	E01	High-pressure switch is disconnected	Check whether there is fault on pressure switch or refrigerant loop
Low-pressure protection	E02	Low-pressure switch is disconnected	Check whether there is fault on pressure switch or refrigerant loop
Water flow protection	E03	There is no or little water in water system	Check water flow , and whether water system is damaged
Antifreezing protection	E07	Flow is inadequate	Check water flow , and whether water system is damaged
Winter Level I antifreezing protection	E19	Ambient temperature is too low	
Winter Level II antifreezing protection	E29	Ambient temperature is too low	
Exhaust over heat protection	P181	Compressor load is too large	Check whether system compressor is in normal operation
AC input undervoltage protection	E21	Input voltage is too low, resulting in too high input current	Check and measure input voltage
AC input overcurrent protection halt	E22	Input current is too high, exceeding effective value for protection current halt	Check and measure input current
DC bus overvoltage protection	E24	DC bus voltage＞DC bus overvoltage halt protection value	Check and measure bus voltage
DC bus undervoltage protection	E25	DC bus voltage＜DC bus overvoltage halt protection value	Check and measure bus voltage
IPM overcurrent protection	E27	IPM current is too large	Check and measure current as well as adjust
IPM module overheat protection	E28	Temperature of IPM module is too high	Check and measure current as well as adjust
Radiator temperature sensor error	E29	Sensor is in short circuit or open circuit	Check and replace sensor
Power component overheat protection halt	E30	Temperature of power component is too high	Reduce temperature and check component
IPM current sampling	E32	Fault of current sampling	Check and measure current as well as adjust
Compressor overcurrent protection hault	E33	Compressor current is too high	Check and measure current as well as adjust
Input voltage default phase	E35	Input voltage is in default phase	Check and measure voltage as well as adjust
Failure of compressor booting	E36	Default phase, step out or hardware damage on actuator	Check and measure voltage and check frequency board hardware
Fault of communication between DSP and communication board	E37	Fault of DSP and communication board	Check communication connection
Fault of communication between DSP and PFC	E38	Fault of DSP and PFC	Check communication connection
Input voltage sampling	E39	Fault of current sampling	Check whether chip is damaged and replace chip
Fault of EEPROM	E40	Chip error	
Compressor flux weakening protection alarm	E41	Inadequate magnetic force of compressor	
Fault of communication between actuator and Pc8002	EE8	Abnormal communication between remote controller and mainboard	Check communication connection

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Form of Troubleshooting

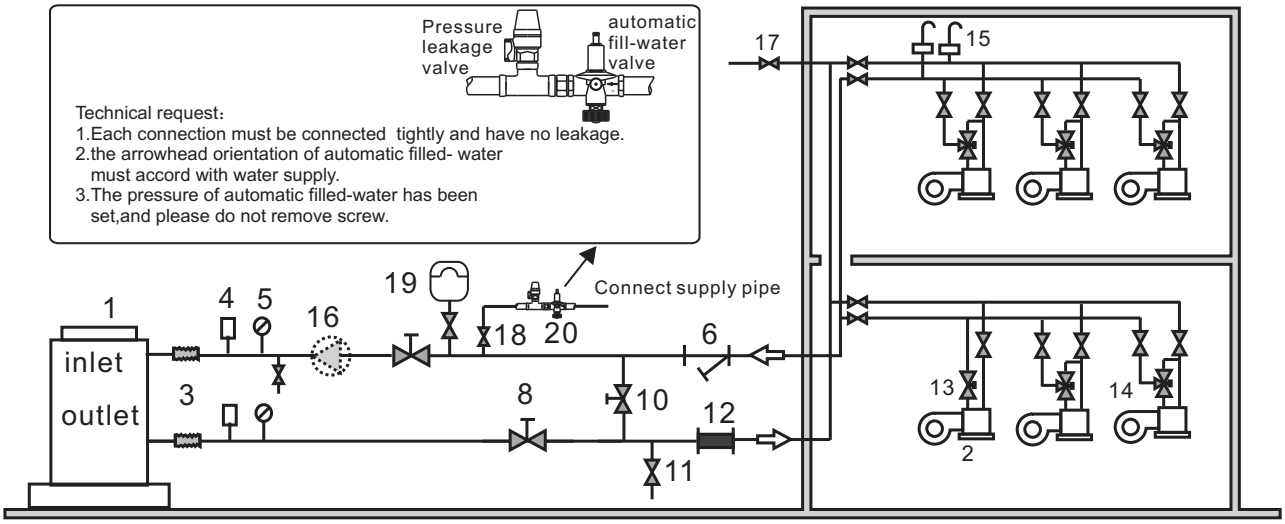
B. Or make judgment and troubleshooting by reference to the following form

Failure	Possible causes for the failure	Solutions
Heat pump cannot be started	1 Wrong power supply 2 power supply cable loose 3 circuit breaker open	1 shut off the power and check power supply; 2 check power cable and make right connection 3 check for the cause and replace the fuse or circuit breaker
Water pump is running with high noise or without water	1 lack of water in the piping 2 much air in the water loop 3 water vavles closed 4 dirt and block on the water filter	1 check the water supply and charge water to the piping; 2 discharge the air in the water loop; 3 open the valves in water loop; 4 clean the water filter.
Heat pump capacity is low, compressor do not stop	1 lack of refrigerant; 2 bad insulation on water pipe; 3 low heat exchange rate on air side exchanger; 4 lack of water flow	1 check for the gas leakage and recharge the refrigerant; 2 make good insulation on water pipe; 3 clean the air side heat exchanger; 4 clean the water filter
High compressor exhaust	1 too much refrigerant 2 low heat exchange rate on air side exchanger	1 discharge the redundant gas 2 clean the air side heat exchanger
Low pressure problem of the system	1 lack of gas 2 block on filter or capillary 3 lack of water flow	1 check the gas leakage and recharge freon; 2 replace filter or capillary; 3 clean the water filter and discharge the air in water loop.
Compressor do not run	1 power supply failure 2 compressor contactor broken 3 power cable loose 4 protection on compressor 5 wrong setting on return water temp. 6 lack of water flow	1 check off the power supply; 2 replace compressor contactor; 3 tighten the power cable; 4 check the compressor exhaust temp.; 5 reset the return water temp.; 6 clean the water filter and discharge the air in water loop.
High noise of compressor	1 liquid refrigerant goes into compressor 2 compressor failure	1 bad evaporation, check the cause for bad evaporation and get rid of this; 2 use new compressor;
Fan do not run	1 failure on fan relay 2 fan motor broken	1 replace the fan relay; 2 replace fan motor.
The compressor runs but heat pump has not heating or cooling capacity	1 no gas in the heat pump; 2 heat exchanger broken; 3 compressor failure.	1 check system leakage and recharge refrigerant; 2 find out the cause and replace the heat exchanger; 3 replace compressor.
Low outlet water temperature	1 low water flow rate; 2 low setting for the desired water temp.;	1 clean the water filter and discharge the air in water loop. 2 reset the desired water temperature.
Low water flow protection	1 lack of water in the system; 2 failure on flow switch	1 clean the water filter and discharge the air in water loop. 2 replace the flow switch.

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Appendix 1 Install sketch map

Especial installation（ expandable water tank）



Remark

1 main unit
2 fan coil
3 rubber flexible connection
4 thermometer
5 manometer
6 Y type filter
8 ball valve
10 bypass valve
11 drain valve
12 filter
13 two-way valve
14 three-way valve
15 automatic ventilation
16 water pump
17 ball valve
18 ball valve
19 expandable water tank
20 automatically filled-water

Installation request:

1 The factory only offers main unit (1)in the legend, and the other modules which are indispensable fittings, are provided by users or installation company.
2 The unit which of code contains the letter "B" ,has water pump inside and need not install water pump outside （16）
3 Automatic ventilation（15）is installed on the top point of the water system。
4 The quantity proportion of two-way valve（13）and three-way valve（14）is referred to the technical regulation, and there is three-way valve installed on the farthest place of water system.
5 The ball valve (17) is used when it is swashed, filled water in the water system and so on.

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Appendix 2：
The installation explanation of automatic filled-water

1 When automatic filled-water valve is installed,the arrowhead orientation of inlet water must accord with the orientation of valve ;

2 Automatic filled-water has been adjusted in advance to 1.5bar；

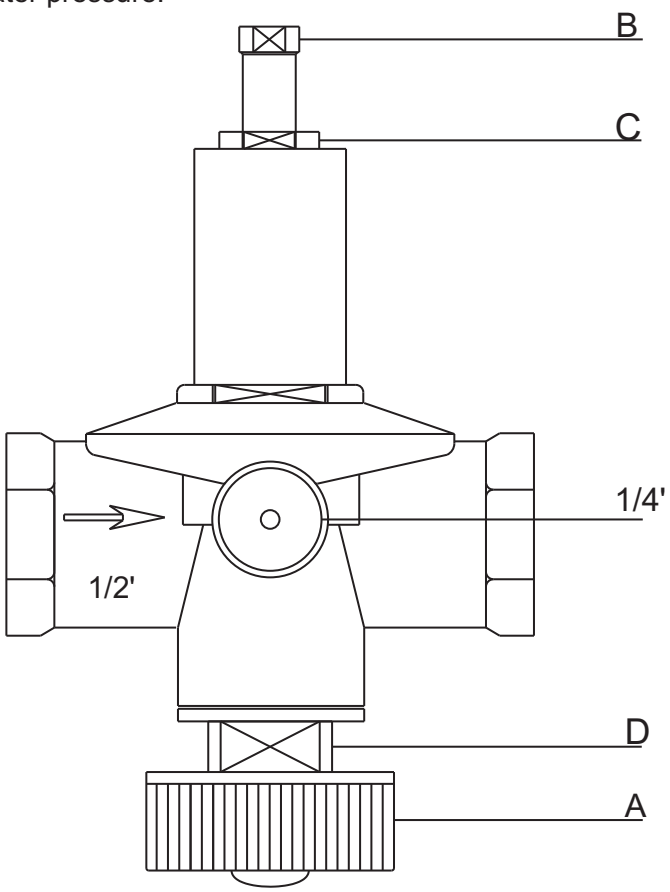
3 If readjust the pressure of inlet water,please operate as follows：

* open the screw cap（C）；
* If reduce the pressure of water supply,pease unscrew the pressure to adjust the screw(B)；
* If increese the pressure of waer supply,please screw down the pressure to adjust the screw (B)

4 When the system need fill water at first,wrest the handle(A) of filled-water.Then the handle(A) can return(close) when the system is full of water.

5 Automatic filled-water Valve need clean in a periodic time and then you must close the tap, unscrew the plug(D),remove the inside filter net.Please assemble them again after cleaning.

NOTICE：There are two connections for water pressure meter in the central section of automatic filled-water,where the water pressure meter can be connected directly and display the set pressure.The screw cap(C) must be tweaked after adjusting the filled-water pressure.



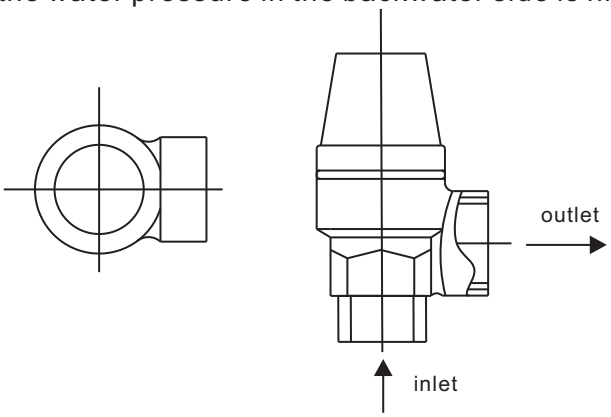
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Appendix 3:

The installation explanation of the leakage pressure valve.

1 The action pressure of leakage pressure valve 'is more than 3bar(valve is open), but the pressure can not be adjusted.

2 The valve will open automatically to make sure that the water loop of air-con system is safe when the water pressure in the backwater side is higher than the set pressure.



Appendix 4:

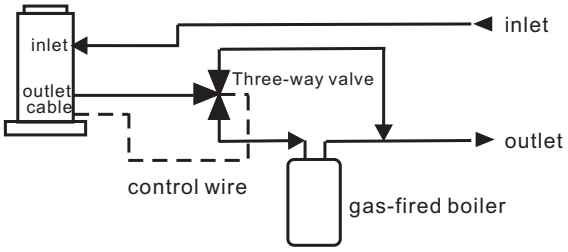
The way of assistant heat source connection

Unit provides the connection of assistant heat source which can not be only for gas-fired boiler,but also for electronic boiler or warm-net pipe for city accordingly.

The way to the connection is as follows:

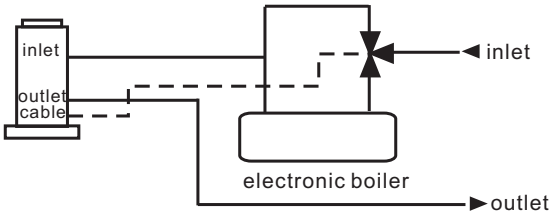
1）water chiller and heat pump+assistant gas-fired boiler

water chiller and heat pump



2）water chiller and heat pump+assistant electronic boiler

water chiller and heat pump

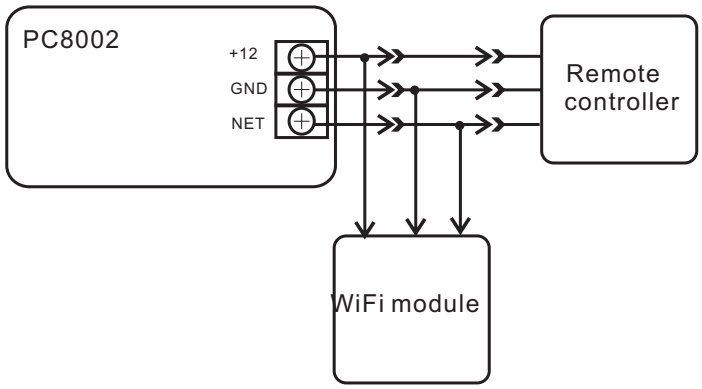


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Appendix 5、Ways of connection to remote controller

The following ways of connection to control system are optional based on user demand:

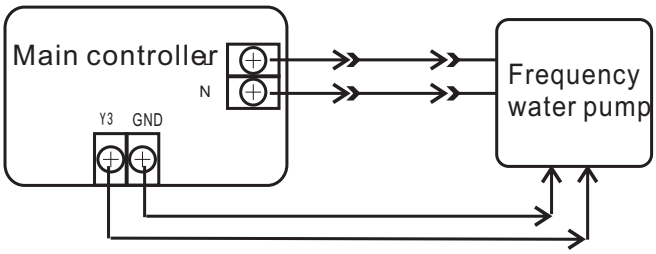
A. Simultaneously connect remote controller and networking interface:



Note:
1、In the mode of unit operation, it is necessary to complete value setting by remote controller.
2、It is advised to control equipment by mobile terminal after connection to WiFi module.

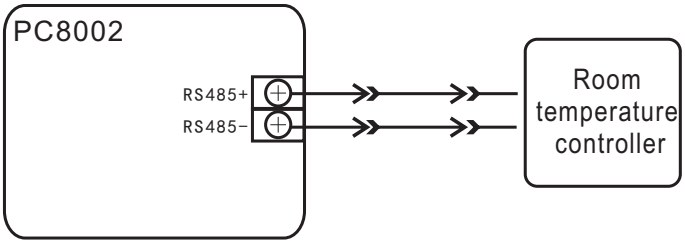
B. Connect with external water pump:

Note:
Connect frequency water pump (PWM control) to reserved interface as required and controller will be able to automatically control water pump.



C. Connect to room temperature controller:

Note:
1、The way is applicable to special requirements of users;
2、Connect 485 signal wire to reserved terminal after determining the use of room temperature controller.



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Appendix 6、Caution & Warning

1. The unit can only be repaired by qualified installer centre personnel or an authorised dealer.（for Europe market）
2. This appliance is not intended for use by persons (including children) with reduced physical sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety.（for Europe market）
   Children should be supervised to ensure that they do not play with the appliance.
3. Please make sure that the unit and power connection have good earthing, otherwise may cause electrical shock.
4. If the supply cord is damaged, it must be replaced by the manufacturer or our service agent or similarly qualified person in order to avoid a hazard.
5. Directive 2002/96/EC (WEEE):
   The symbol depicting a crossed-out waste bin that is underneath the appliance indicates that this product, at the end of its useful life, must be handled separately from domestic waste, must be taken to a recycling centre for electric and electronic devices or handed back to the dealer when purchasing an equivalent appliance.
6. Directive 2002/95/EC (RoHs): This product is compliant with directive 2002/95/EC (RoHs) concerning restrictions for the use of harmful substances in electric and electronic devices.
7. The unit CANNOT be installed near the flammable gas. Once there is any leakage of the gas , fire can be occur.
8. Make sure that there is circuit breaker for the unit, lack of circuit breaker can lead to electrical shock or fire.
9. The heat pump located inside the unit is equipped with an over-load protection system. It does not allow for the unit to start for at least 3 minutes from a previous stoppage.
10. The unit can only be repaired by the qualified personnel of an installer center or an authorized dealer.（for North America market）
11. Installation must be performed in accordance with the NEC/CEC by authorized person only.（for North America market）
12. USE SUPPLY WIRES SUITABLE FOR 75℃.
13. Caution: Single wall heat exchanger, not suitable for potable water connection.

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Appendix 7、Cable specification

1. Single phase unit

Nameplate maximum current	Phase line	Earth line	MCB	Creepage protector	Signal line
No more than 10A	2×1.5mm^2	1.5mm^2	20A	30mA less than 0.1 sec	n×0.5mm^2
10~16A	2×2.5mm^2	2.5mm^2	32A	30mA less than 0.1 sec	
16~25A	2×4mm^2	4mm^2	40A	30mA less than 0.1 sec	
25~32A	2×6mm^2	6mm^2	40A	30mA less than 0.1 sec	
32~40A	2×10mm^2	10mm^2	63A	30mA less than 0.1 sec	
40~63A	2×16mm^2	16mm^2	80A	30mA less than 0.1 sec	
63~75A	2×25mm^2	25mm^2	100A	30mA less than 0.1 sec	
75~101A	2×25mm^2	25mm^2	125A	30mA less than 0.1 sec	
101~123A	2×35mm^2	35mm^2	160A	30mA less than 0.1 sec	
123~148A	2×50mm^2	50mm^2	225A	30mA less than 0.1 sec	
148~186A	2×70mm^2	70mm^2	250A	30mA less than 0.1 sec	
186~224A	2×95mm^2	95mm^2	280A	30mA less than 0.1 sec	

2. Three phase unit

Nameplate maximum current	Phase line	Earth line	MCB	Creepage protector	Signal line
No more than 10A	3×1.5mm^2	1.5mm^2	20A	30mA less than 0.1 sec	n×0.5mm^2
10~16A	3×2.5mm^2	2.5mm^2	32A	30mA less than 0.1 sec	
16~25A	3×4mm^2	4mm^2	40A	30mA less than 0.1 sec	
25~32A	3×6mm^2	6mm^2	40A	30mA less than 0.1 sec	
32~40A	3×10mm^2	10mm^2	63A	30mA less than 0.1 sec	
40~63A	3×16mm^2	16mm^2	80A	30mA less than 0.1 sec	
63~75A	3×25mm^2	25mm^2	100A	30mA less than 0.1 sec	
75~101A	3×25mm^2	25mm^2	125A	30mA less than 0.1 sec	
101~123A	3×35mm^2	35mm^2	160A	30mA less than 0.1 sec	
123~148A	3×50mm^2	50mm^2	225A	30mA less than 0.1 sec	
148~186A	3×70mm^2	70mm^2	250A	30mA less than 0.1 sec	
186~224A	3×95mm^2	95mm^2	280A	30mA less than 0.1 sec	

When the unit will be installed at outdoor, please use the cable which can against UV.

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